Introduction

Priority Queue
Data Structures and its Applications (UE25CS52A)

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Quiz : Priority Queue

1. **. Which statement best describes a priority queue ADT?**

a) Elements are always processed in FIFO order
b) Elements are processed according to their associated priorities
c) Elements are processed in LIFO order
d) Elements are stored only in sorted order

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Quiz : Priority Queue

2. In a max-priority queue, which element is removed first

a) One with the highest priority which came later
b) One with the highest priority which came earlier
c) Any one with the highest priority
d) None of these

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Quiz : Priority Queue

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Quiz : Priority Queue

3. **Which statement correctly distinguishes a priority queue from an ordinary queue?**

a) A priority queue allows insertion but not deletion

b) An ordinary queue always uses priorities

c) A priority queue determines deletion primarily by priority

d) An ordinary queue cannot be implemented using an array

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Quiz : Priority Queue

4. Which statement about equal priorities is correct?

a) Equal-priority elements are not allowed
b) All should be selected together
c) The queue must always use FIFO for equal priorities
d) The queue must always use LIFO for equal priorities

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Quiz : Priority Queue

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Quiz : Priority Queue

5. Which algorithm may not use priority queue?

a) Patient management
b) Scheduling processes in an operaring system
c) Managing function calls
d) Selecting the next print job from the spool

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Quiz : Priority Queue

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Priority Queue: concept

- A priority queue is an abstract data type in which each element is associated with a priority.
- A priority queue removes the component with the highest priority first.
- A priority is a number associated with each element
- Higher the numeric value, could mean higher the priority.
- Or lower the numeric value, could mean higher priority

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Priority Queue: some examples

- Patient treatment
- Scheduling of processes

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Priority Queue: interface

initialize() deinitialize()	Create an empty queue Make the queue empty
enqueue(x) I	Insert x We may add in number of ways. This in turn would affect dequeue
dequeue()	Remove the element with the highest priority. This in turn depends how enqueue operation was done
isEmpty()	Check whether the queue is empty
isFull()	Check whether the queue is full

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Priority Queue: Possible Implementations

- Unodered list
- Ordered list : ordered based on priority
- Unordered array
- Ordered array : ordered based on priority
- Heap tree

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Priority Queue: Applications of priority queue :

- CPU Scheduling of processes
- Hospital Emergency Services
- Printer Scheduling
- Network Packet Scheduling
- Dijkstra's Shortest Path Algorithm
- Prim's Minimum Spanning Tree Algorithm
- Event-Driven Simulation
- Huffman Coding

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THANK YOU

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